

Master thesis

Master of Science (M.Sc.)
Department of Business
Major: Data Science

Update the
title page
with related
text

Topic:

**Enhancing Solar-Induced Fluorescence OCO-2 data
for High-Resolution Crop Yield Prediction in
Germany**

Author: VT

Matriculation Number:
First supervisor: Prof. Dr.
Second supervisor: Mr.
Submitted on: 20.08.2026

Statutory Declaration:

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Berlin, xx.xx.202x

.....

(Signature)

[your name]

Abstract

Four sentences of background and importance of this topic.

Four sentences defining the gap analysis (work not done in field) and your problem statement.

Four sentences about the methodology of your work.

Six sentences of Results and significance of your results.

Two sentence summarizing your contribution.

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write a one page abstract
- Approximately 20 sentences with importance, background, current state, gap analysis, problem statement, your solution, your methods, your results, and improvement in field being discussed.

Zusammenfassung

Da die meisten Leuten an der TU deutsch als Muttersprache haben, empfiehlt es sich, das Abstract zusätzlich auch in deutsch zu schreiben. Man kann es auch nur auf deutsch schreiben und anschließend einem Englisch-Muttersprachler zur Übersetzung geben.

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1 Introduction

One-Two paragraph(s) introducing the purpose and theme of the research.

1.1 Motivation

Two-Four paragraph(s).

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1.2 Objective

One-Two paragraph(s).

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'Figure 1.1 shows Component X as part of ...'

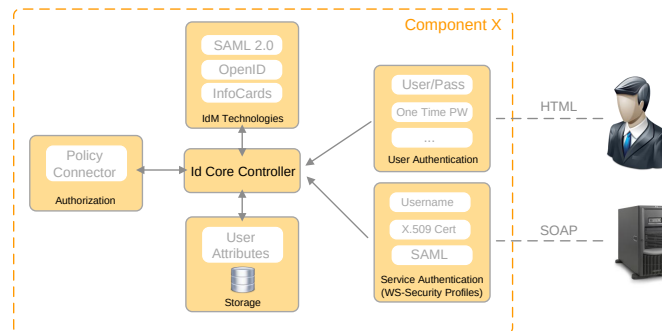


Figure 1.1: Component X

1.3 Thesis Scope

One-Two paragraph(s).

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1.4 Purpose and goal

One-Two paragraph(s).

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1.5 Outline

The 'structure' or 'outline' section gives a brief introduction into the main chapters of your work. Write 2-5 lines about each chapter. Usually master thesis is separated into 6-8 main chapters.

This example thesis is separated into 7 chapters.

Chapter 2 analyzes the requirements for your component. This chapter provides the reader the background knowledge to understand technical concepts applied in the thesis. This chapter will have 5-10 pages.

Chapter 3 is usually termed 'Related Work', 'State of the Art'. What did other scientists propose regarding your topic? This chapter makes about 20 percent of the complete thesis.

Chapter 4 is termed 'Research question'. Here you describe your Problem statement and Research questions.

Chapter 5 describes the methodology/implementation part of your work. Don't explain every code detail but emphasize important aspects of your implementation. This chapter will have a volume of 20-30 percent of your thesis.

Chapter 6 is Experiments, Results and Discussion. This chapter will have a volume of 15-20 percent of your thesis.

Chapter 7 summarizes the thesis, describes the problems that occurred and gives an outlook about future work. Should have about 4-6 pages.

2 Theoretical Background

This chapter provides the reader the background knowledge to understand technical concepts applied in the thesis.

2.1 Background

2.1.1 Definitions

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2.2 Existing Methods

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2019.

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2.3 Evaluation Metrics

2.3.1 Subsection 1

2.3.2 Subsection 2

2.3.3 Subsection 3

3 Related work

In this chapter, we discuss the relevant literature review for the chosen topic.

Identify the important subcomponents (e.g., in terms of methods, evaluation, preprocessing or visualization) of your thesis and for each subcomponent, make a separate section with subsections as shown below.

The summary of the literature is shown in Table ??.

3.1 Section 1

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2023.

Group in terms of similarity in approach.

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3.2 Section 2

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Group in terms of similarity in approach.

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4 Research question

This chapter introduces the research question ...

4.1 Problem statement

4.2 Research question

5 Methods

5.1 Research Design and Study Area

5.1.1 Overall Research Design

The thesis study followed a predictive research design that links satellite based SIF enhancement with an agricultural yield application. Rather than designing the work as a single end to end model, it was modeled in two analytical stages. The first stage uses machine learning based models to learn relationships between OCO-2 SIF observations and spatially complete biophysical, radiation, seasonal and crop-type predictors. The second stage uses the enhanced SIF data produced by the best performing model; Spatial Aggregate (4km) restricted to winter wheat pixels which was then grouped and summarized by month and NUTS level 3 administrative regions and finally used as a predictor for the winter-wheat yield model. To reiterate, SIF data was first spatially enhanced, given completeness spatially and temporally and this enhanced SIF data is finally used to test if it could provide useful signal for a crop yield prediction model. Both of the stages used historical predictor data with held-out test or inference data which ensured the results describe a predictive association between the target and predictor variables rather than a casual effect of an individual predictor.

The SIF enhancement stage was designed as a supervised mapping between a set of spatial predictors to a SIF value with a defined spatial support (spatial resolution of the predictors). We can define this mathematically as

$$\hat{S}_{u,t} = f_{\theta}(\mathbf{X}_{u,t}), \quad (5.1)$$

where f_{θ} represents a fitted tree-based, multilayer-perceptron, or CNN model with parameters θ , u denotes the spatial support or OCO-2 footprint, t the observation time, $\mathbf{X}_{u,t}$ the predictor variables, and $S_{u,t}$ the OCO-2-derived target (this input target representations can vary from a single data point to spatial aggregates combined multiple targets into one value, this will be discussed in depth in ??). For area-to-point MLP, $\mathbf{X}_{u,t}$ was a footprint-level matrix containing the predictor values for every 20 m pixel, For polygon-mean models, $\mathbf{X}_{u,t}$ was a tabular summary of footprint-level statistics, For CNN models, $\mathbf{X}_{u,t}$ was a multi-channel raster chip.

The SIF enhancement experiments used different model architectures and data representations (both target and predictors). For the polygon-mean models, mean aggregation of predictor values falling within an OCO-2 footprint were averaged and fitted to a single SIF value, Area-to-point model uses a multilayer perceptron to fit pixel level sentinel-2 data to footprint scale SIF. Single footprint CNN experiments uses one target footprint in one image chip (channels representing predictors), whereas multi footprint CNN experiments uses multiple same date footprints in one image chip after which a footprint mask-averaged prediction was computed. The best experiment; spatial-aggregate (4km) combines by averaging at least 5 OCO-2 observations in a fixed 4km grid cell and a CNN model is trained on it with an aggregate footprint-weight map as many of the 4km grid cells will have some it's SIF polygons falling partially outside the cell. Given that almost all of the experiments have different architectures and data representations, if one model performs better, the improvement cannot be attributed only to it being a CNN, MLP,

Random Forest, or XGBoost model. The result reflects the entire combination of data preparation, predictors, target definition, spatial support, and model architecture.

The downstream task of predicting winter wheat yield used the same predictive logic as the first task, but works on a much more coarser spatial scale. Since the yield data available is at NUTS 3 level which is the highest resolution, we can only conduct the yield analysis at this spatial scale. For NUTS-3 region r and year y , the winter-wheat yield can be represented as

$$\hat{Y}_{r,y} = g_{\phi}(S_{r,y,3}, S_{r,y,4}, S_{r,y,5}, S_{r,y,6}, S_{r,y,7}), \quad (5.2)$$

where $S_{r,y,m}$ denotes a monthly SIF predictor for March through July and g_{ϕ} denotes the fitted yield model. To see if the SIF enhancement helps improve the yield modelling, separate versions of the input data were produced in the form of crop pure enhanced SIF (ie; only winter wheat pixels' SIF values were taken), raw OCO-2 footprint SIF values which is a representation of mixed crop SIF values. The raw OCO-2 SIF values was also constrained to SIF polygons containing at least 10% winter wheat out of the SIF footprint crop area. The yield model design held the yield target, region-year groupings and Random Forest configuration constant while changing the SIF representation. To reiterate, the crop pure predictors where SIF values taken from crop pixels that were winter wheat, whereas the raw baselines took OCO-2 footprints with either no crop purity requirement or a minimum crop (winter wheat) coverage threshold (5%, 10%, 30%). Years 2019-2022 subset was used for model training and 2024 subset was used as the test data. This ensured any performance evaluation can be said be generalized over years, ie; it measured whether crop-targeted enhancement reduced out-of-year prediction error compared to SIF observations that were combination of different crop SIF values for a given footprint.

5.1.2 Geographical Scope

The SIF enhancement study domain was Germany, where the data domain was represented by a set of Sentinel-2 Military Grid Reference System (MGRS) tiles rather than by a single national level raster. Since the Sentinel-2 data (the main high resolution predictor data) was only available in MGRS tiles format, we have to stick to tile based data domain rather than state boundary, agricultural zone etc. based domain. The nine MGRS tiles used for the Sentinel-2 training predictor data were T32ULA, T32ULB, T32UMD, T32UNA, T32UNC, T32UNE, T32UPC, T32UPE, and T32UPU. These tiles sampled data across north, centre and south of Germany and included all the major crop-producing states in Germany: Bayern, Niedersachsen, Sachsen-Anhalt, Mecklenburg-Vorpommern, Nordrhein-Westfalen, Rhinland-Pfalz, Hessen. Figure ?? shows the distribution of SIF polygons from 2019-2024 in the 9 MGRS tiles. Since the experiment design has different model architectures and data representations, we cannot claim an uninterrupted high resolution coverage of every location in Germany for every experiment as the predictions are restricted by availability of OCO-2 tracks, Sentinel-2 product availability, quality filters and selected months in the input data. Therefore any Germany-wide conclusions can only refer to the diversity represented by the selected tiles and observations rather than to an exhaustive census of all German pixels.

Two additional MGRS tiles, T32UQV and T32UMC, were used for external native-footprint evaluation of the best model; Spatial-aggregate (4km) model. This helped quantify if target to predictor relationships learned on the 9 MGRS tiles could be transferred to new locations or in this case, new MGRS tiles. Also the external native-footprint evaluation tests whether a 20m resolution SIF map predicted by the model and then aggregated to the native OCO-2 footprint matched with the original OCO-2 footprint SIF value, if it

tested well then we can conclude that both geographic transfer to unseen tiles and transfer from aggregated training support back to approximately native OCO-2 footprint support works well.

The winter-wheat yield model focused on Bavaria as it is the largest wheat producing state in Germany and making 20m resolution SIF maps was extremely computationally and time consuming. Fives MGRS tiles were used for the winter-wheat yield model in Bavaria: T32UNA, T32UPU, T32UQV, T32UPA, and T32UPV. A Bavarian NUTS 3 region was retained if at least 50% of it's area fell within one of the above MGRS tiles. Visual inspection of the NUTS 3 regions as seen in Figure ?? and associated winter wheat mask map as seen in Figure ?? helped confirm that the NUTS 3 regions that were only partially present in the MGRS tile had majority of their wheat producing areas in the MGRS tile. Yield prediction was done at NUTS 3 region-year level because this was the highest spatial and temporal availability of the official yield data.

Agro-climatic variation in the form of Plant Hardiness zones was represented independently of the administrative and raster tiling data domains. Each SIF observation inherited one or more plant hardiness zone labels from the spatial zone layer, and model performance was later summarized for the individual zones 7a, 7b, 8a, 8b, and 9a. Mixed labels occurred due to some of the SIF polygons falling across multiple plant hardiness zones. MGRS tiles, federal states, NUTS regions, and hardiness zones therefore served different purposes and were not treated as interchangeable regional definitions, each are suited to how the data was available. MGRS tiles defined Sentinel-2 raster storage and experimental partitions, federal states gave geographic context, NUTS 3 regions was required for the yield target support and plant hardiness zones described agro-climatic conditions. Keep these geographic roles separate allowed a fitted SIF enhancement model to be evaluated by state boundary geography, season, agro-climatic zone without changing its corresponding prediction target.

5.1.3 Temporal Scope and Spatial Supports

The main SIF enhancement predictor and target datasets covering year range 2019-2024 focused on months February to July as these are the main growing period for Winter Wheat in Germany. OCO-2 observations are basically tied to individual acquisition dates, the OCO-2 satellite do not make regular revisit over the same track instead are staggered by weeks to months around the track and therefore highly irregular in time. Sentinel-2 predictors are monthly composites whose composite temporal range can vary from 7 days to 45 days. GLASS products other than GLASS PAR are 8-day composites, eg; Day of Year 01 composite (DOY 01) covers DOY 01 to DOY 08. GLASS PAR is available as daily source files. VIIRS PAR is represented as daily mean source files. Dates were matched according to the available date range of each product rather than with one common nearest date rule. The predictor variables were temporally matched with the OCO-2 observation dates in the following method: Sentinel-2 records were linked by month and product interval, eight-day GLASS products were linked to the interval containing the SIF date, and VIIRS PAR was matched by exact date as it is a daily source file. OCO-2 records that were outside a Sentinel-2 product interval as obtained from the supporting WASP DTS raster files (these files were available for 10m and 20m reflectance bands and contained days since start of year for each pixel, thus essentially allowing us to identify the temporal range of the monthly composite) were retained with a `date_align` flag for later evaluation. The yield analysis used March to July in 2019-2022 and 2024, while 2023 was excluded because Sentinel-2 WASP products had poor coverage for this year. Years 2019-2022 were used for model development and 2024 was held out for testing. The predictors were kept as separate month wise average predictors allowing to preserve any seasonal information in

winter wheat growth and senescence.

The thesis study's experiments developed several spatial supports (target and predictor spatial resolution) that were kept distinct for each modelling and evaluation experiment. Sentinel-2 bands at 10 m and 20 m were aligned to a common 20 m grid to better align spectral index calculation and also reduce computational complexity. Every other data source would later be aligned to this common 20m grid as in the case of the CNN models since they require consistent image resolutions and dimensions for training. The 10m crop type rasters were converted to fractional crop maps on the common 20m grid. GLASS FAPAR had an actual pixel spacing (resolution) of approximately 231.7 m while VIIRS PAR had a spacing of approximately 927 m, these were both aligned with the Sentinel-2 predictors. Each 20m pixel would acquire its corresponding GLASS predictor value depending on the alignment, many 20m pixels can have the same GLASS values due to the considerable resolution differences. OCO-2 observations were irregular polygons ranging from elongated parallelograms shape (Nadir) to triangle, trapezoid shape (Glint) with areas typically close to 1 km². Single-footprint and multi-footprint enhancement experiments preserved the individual footprint targets, but as the name suggests, multi-footprint chips included several separate masks (each mask corresponding to a SIF footprint). Spatial-aggregate (4km) experiment averaged at least five same-date observations within a fixed 4 km cell represented by 200x200 predictor pixels (each pixel at 20m resolution). External Native OCO-2 resolution inference tested the Spatial-aggregate (4km) model by averaging its 20 m predictions over individual OCO-2 footprints in unseen tiles and comparing the averaged value to the original SIF observation value. For the winter wheat yield model, Spatial-aggregate (4km) model predictions over each 10m winter wheat pixels (regridded to 20m) were aggregated to one monthly value for each NUTS-3 region and year. The Spatial-aggregate (4km) model produces SIF predictions at 20 m, but since there is no real 20m SIF data for validation, the validation could only be done at Native OCO-2 SIF resolution (around 1 km²). The 20m SIF predicted map is thus a downscaled estimate and not a directly verified 20m ground truth, more will be discussed in the Results section.

5.2 Datasets

5.2.1 OCO-2 SIF Observations

NASA OCO-2 observations provided the target variable for all the SIF enhancement experiments. The data source was the Level 2 bias-corrected solar-induced fluorescence product generated with the IMAP-DOAS retrieval and distributed as daily files planet wide coverage by the NASA Godard Earth Sciences Data and Information Services Center (GES DISC). Two product releases V11.r and V11.2r were combined row-wise and used for the thesis study [OPC20, OPC24]. The files contained daily-corrected SIF retrievals at 740 nm, 757 nm, and 771 nm, together with their retrieval-uncertainty fields, acquisition time, latitude and longitude, four latitude & longitude corners describing the elongated ground footprint (usually in the form of an elongated parallelogram), quality flags, measurement mode, viewing geometry, illumination geometry, source-file identifiers, and selected meteorological fields. Measurement mode determined if the observation was retrieved in Nadir or Glint mode. Quality Flag determined the quality of the SIF observations with higher number flag given to observations obscured by cloud, haze and other climate conditions, consequently Quality Flag (0,1) was chosen for this study. Phase angle was calculated from the viewing geometry and illumination geometry and helped determine if SIF values drastically changed in a given area if the phase angle changed considerably. The temporal range for OCO-2 observations for this study covered 2019-2024. Even though years preceding and following this range existed, due to the temporal availability of year wise crop

type rasters (2019-2024), the study's data temporal range was restricted to 2019-2024.

5.2.2 Sentinel-2 L3A WASP Monthly Composites

The driving predictor data for high resolution SIF enhancement was Sentinel-2 L3A WASP data, these were monthly composites Sentinel-2 composites produced by the Weighted Average Synthesis Processor (WASP). WASP forms temporal syntheses from atmospherically corrected Sentinel-2 Level 2A inputs generated with MAJA, with the processing specifications documented by Hagolle et al. [HHD⁺17, HMK18]. The WASP products (product referring to one year-month composite data directory containing the various Band raster and supporting files) were organized by MGRS tile, year and monthly synthesis date which provided consistent spatial reference for the study areas in Germany. The monthly composite product supplied blue, green, red, and near-infrared reflectance bands B2, B3, B4, and B8 at 10 m resolution. Red-edge, narrow near-infrared, and short-wave-infrared bands B5, B6, B7, B8A, B11, and B12 were available at 20 m resolution. Each product also contained 10 m and 20 m classification layers, identified as `FLG_R1` and `FLG_R2`, alongside the Band rasters which helped determine if the data pixel was land, water, cloud, snow or no data. Band (reflectance) value was encoded as signed 16-bit integers with a quantification value of 10,000, with no data assigned as -10,000. The temporal coverage for the data was monthly composites for February to July during 2019-2024. Since 20m resolution was still a high resolution, 10m reflectance bands were aggregated to 20m resolution in the data preprocessing steps since all the input data needed to be at the same spatial resolution and a lot of computation power was gained with not dealing with 10m resolution. Each year-month composite collection was around 30-35GB in size and took around 30-50mins to download from Centre national d'études spatiales (CNES) GEODES portal. An earlier WASP collection from Deutsches Zentrum für Luft und Raumfahrt (DLR) EOC service was available for download, but data preceding 2024 were archived in May, 2026 and was no longer available for download.

5.2.3 GLASS Vegetation and Radiation Products

The Global LAnd Surface Satellite (GLASS) product suite provided medium resolution vegetation index and radiation variables. Regular updates and additions of new spatial data to the GLASS product suite have been done from 2013-2023 as shown in [LZL⁺13, LZX⁺13, LCJ⁺21, LCC⁺23]. Before an expensive model data preparation and model training could be done on very high resolution Sentinel-2 predictors (20m resolution), the GLASS provided predictors was used to test if the overall data preparation methodology and model architectures were suited for the SIF enhancement task. The GLASS products were fraction of absorbed photosynthetically active radiation (FAPAR), photosynthetically active radiation (PAR), normalized difference vegetation index (NDVI), and enhanced vegetation index (EVI). FAPAR was obtained from the GLASS09D01 V60 archive at 500m resolution, PAR was obtained from GLASS04B01 V42 at 0.05° spatial resolution while NDVI and EVI were obtained from the corresponding GLASS13D01 and GLASS14D01 250 m MODIS products. These products other than PAR were supplied as eight-day composites on the MODIS sinusoidal grid and stored in HDF format. MODIS sinusoidal tiles `h18v03` and `h18v04` were the two tiles that covered the entirety of Germany. The temporal range was 2019-2024 and focused on composite start days from day of year 033 to 209 (covering February to July approximately). PAR in contrast was supplied as daily source files, but for the case of this study, eight day sequence matching FAPAR, NDVI and EVI day of year were downloaded to reduce the amount of data to be downloaded and processed. These GLASS products will also be referred to as Moderate Resolution

Imaging Spectroradiometer (MODIS) variables as they are enhanced data derived from raw spectroradiometer data as given in FILLREF.

5.2.4 VIIRS Daily PAR Product

Daily Photosynthetically Active Radiation (PAR) data for the principal Sentinel-2 workflow came from the VIIRS/NPP Photosynthetically Active Radiation Daily Level 3 Global 1 km Sinusoidal Grid Version 2 product, identified as VNP18A2 [Wan25]. This data has a much higher resolution (1km) than the GLASS provided PAR data at 0.05° . The product provides daily average photosynthetically active radiation and its associated categorical quality layer. Its native grid is a VIIRS sinusoidal projection with a pixel resolution of approximately 927m. This sinusoidal project or coordinate reference system was later realigned with the optical predictor grid as described in Section 5.3. The quality layer provides quality codes that describe the data's usability rather than a continuous numerical ranking between the retrievals. Instantaneous PAR was also available at every 3 hours intervals daily, but Daily-mean PAR was chosen instead because the OCO-2 SIF values were daily-corrected SIF values. Rasters in the form of HDF files were downloaded for the unique OCO-2 observation dates.

5.2.5 Crop, Regional, and Yield Data

Annual crop type raster was obtained from the German Aerospace Center (DLR) EOC Geoservice [AGAG⁺22, GHA⁺25]. The yearly rasters provided a 10m resolution categorical raster that marked and labelled every classified crop pixel on Germany's map into 21 distinctive crops falling in the crop groups: cereal, oilseed, root-crop, grassland, woody-crop, other agricultural, and non-crop classes. These annual crop type rasters provided the crop spatial information needed for footprint level crop composition predictor variables and downstream winter wheat yield analysis. Administrative boundaries of Germany, German federal state boundaries, NUTS 3 regions were obtained from the Geographic Information System of the Commission through the `giscoR` interface [Her26]. NUTS level 1 represented German federal states, while NUTS level 3 represented the districts and independent cities used in the Bavarian yield analysis. German plant hardiness zone polygons defining the agro-climatic regions for Germany were acquired by downloading the geojsons in the leaflet data directory provided by [Pla26]. Official Bavarian crop harvest tables provided district-level yields for 2019, 2020, 2021, 2022, and 2024 [Bay26]. Year 2023 was not included in the yield model study because Sentinel-2 WASP products had limited temporal coverage for 2023, ie; not all the necessary months were available for all previously defined MGRS tiles. The target variable for the yield model was *Winterweizen (einschl. Dinkel und Einkorn)* which reports winter wheat yield. The yield values were reported in decitonnes per hectare using German numeric formatting, these would be converted to tonnes per hectare to allow for cross-study comparison.

5.3 Data Preprocessing

5.3.1 OCO-2 Target and Footprint Preparation

The downloaded OCO-2 files were converted into a row-wise observation table where each row represented one SIF observation (sounding). The main target variable was constructed from the daily corrected SIF values at 757nm and 771nm. For sounding i , the combined target as noted in FILLREF was defined as

$$S_i = \frac{S_{757,i} + 1.5S_{771,i}}{2}, \quad (5.3)$$

where $S_{757,i}$ and $S_{771,i}$ are the daily-corrected retrievals in the two oxygen absorption bands. 771nm SIF retrieval is multiplied by 1.5 to bring it closer to the scale of 757nm retrieval before we can combine both the values into one SIF value is that much more noise free than the individual retrievals. This combined target is called `target_modis_sif` in the processed SIF data. The 740nm retrieval target was used for the initial experiments, MODIS 16×16, area-to-point as seen in FILLREF. As mentioned earlier, every OCO-2 SIF retrieval comes with an uncertainty factor, retrieval uncertainty was adjusted using the OCO-2 daily correct factor before any unreliable observations could be removed. Let $\sigma_{757,i}$ and $\sigma_{771,i}$ denote the corrected uncertainties, the uncertainty of the combined target was calculated as

$$\sigma_i = \frac{1}{2} \sqrt{\sigma_{757,i}^2 + (1.5\sigma_{771,i})^2}. \quad (5.4)$$

In this study, negative SIF was not automatically removed eventhough theoretical SIF value cannot be negative. This is because small negative retrievals can occur during observation when the signal is close to zero. A negative value was accepted when $S_i + 2\sigma_i \geq 0$ and marked as questionable when $S_i + 2\sigma_i \geq 0$ check failed and $S_i + 3\sigma_i \geq 0$ was satisfied, and finally the negative value was rejected when $S_i + 3\sigma_i < 0$ was satisfied. The main datasets for the experiments kept negative SIF values that were classified as accepted by this rule. After this, a broad SIF bin range check of $-0.5 \leq S_i < 2.0$ was then used to remove implausible combined values. Quality Flag and Measurement mode was restricted to values 0,1. For Quality Flag this represented best and good values, whereas for Measurement mode this represented Nadir mode and Glint mode.

Each SIF sounding (observation, row) was built into a polygon from its four recorded corner coordinates instead of using the existing latitude and longitude, as the latter were found to be offset by some distance from the polygon's true centroid. This would complicate later SIF polygon assignments, alignments to various grids, boundary selections. The corner pairs were ordered from the original matrix representation to form a closed polygon in WGS 84 coordinate reference system (CRS), and any invalid polygons formed during polygon construction were repaired if possible. For this study, polygon area and centroid operations were carried out after transformation to the coordinate system ETRS89-LAEA Europe (EPSG:3035), so that distances and areas were measured in metres as the WGS 84 CRS is for latitude and longitude representation. The centroid retrieved from the polygon was used for assignment to federal states, plant hardiness zones, MGRS tiles, fixed grid cells, and NUTS-3 regions. The polygon geometry was important for raster extraction and for fractional footprint masks. This was important as an elongated OCO-2 footprint can cross a raster or administrative boundary even when its centroid lies on only one side. Every SIF observation row also kept a row identifier, acquisition date, source file. These metadata fields allowed SIF observations to be traced through later joins, chip manifests, model predictions, and any subgroup evaluations.

5.3.2 Temporal Matching and Spatial Alignment

Most of the data products as seen in FILLREF do not have exact temporal alignment or matching, therefore the predictor dates were matched according to the temporal meaning of each data product. An OCO-2 SIF observation was matched to the Sentinel-2 monthly composite for the same tile, year and month. As mentioned earlier, each Sentinel-2 composite comes with its own DTS raster which contains the lower and upper bound

date acquisition ranges of each pixel in a reflectance (band) raster, these lower and upper bound dates were compared with the OCO-2 observation date. SIF observations inside this interval were labelled as **inrange** while observations outside were labelled **outrange**. Both date alignment groups were retained in the main dataset because most of the out-of-range differences were small and their performance could be reported separately FILLREF. The GLASS products eight-day FAPAR, EVI, and NDVI datasets were matched by the composite interval containing the SIF day rather than by the numerically closest composite start date. For example, a SIF observation on day of year DOY07 belongs to the product beginning on DOY01, since DOY01 composite describes days 01-08. Daily VIIRS PAR was matched to the exact OCO-2 date as they were available as daily source files, while the earlier GLASS PAR product used the closest available day to a SIF date.

In regards to spatial alignment, for the CNN models all image channels within one convolutional sample (chip) had to share the same rows, columns, extent, and coordinate reference system. The experiments using Sentinel-2 data thereby used a 20m reference grid in the UTM coordinate system. Sentinel-2 10m reflectance bands were reduced to this 20m grid by area averaging, while reflectance bands already at 20 m did not need readjustments. Each convolutional sample (chip) cropped its own GLASS FAPAR and PAR section and reprojected it to the 20m grid using bilinear interpolation. For VIIRS PAR, the PAR quality mask was applied on the native VIIRS grid before reprojection, and only quality codes 1 and 2 were accepted. For GLASS products, FAPAR values outside the range 0 to 1 and PAR values outside the range 0 to 700 were treated as missing data. APAR was calculated as the pixel-wise product of FAPAR and PAR.

Categorical information had to be handled in a different way compared to the previous continuous reflectance and radiation (FAPAR, PAR, APAR) data. The Sentinel-2 classification masks provided by the FLG rasters were used to keep pixels that were only land recorded values, this was done before any of the spectral indices were calculated. Annual crop codes were converted into separate binary membership rasters and then averaged onto the 20m grid, this would be used the crop channel for the CNN model input chips. For the crop channel a value of 0.75 means that three quarters of the contributing 10m area (croptype rasters are originally at 10m resolution) belonged to the stated crop group. Since OCO-2 SIF polygons were vector data, they were rasterized as fractional masks, so boundary pixels could take values between zero and one according to the proportion of the pixel covered by the SIF polygon. Fractional masks were preferred to binary masks because a 20m (Sentinel-2 experiments) or 250m pixel (modis predictors, for the experiments FILLREF) rarely follows the slanted edge of a SIF polygon exactly, ie; the SIF edge polygon almost always diagonally cuts a pixel.

5.3.3 Predictor and Crop-Channel Construction

This section will present how the various channels for a CNN model input chip (image) were constructed. For this study, five spectral indices were calculated from land masked Sentinel-2 reflectance bands. They are defined as

$$\begin{aligned}
 \text{NDVI} &= \frac{B8 - B4}{B8 + B4}, & \text{NDMI} &= \frac{B8 - B11}{B8 + B11}, \\
 \text{NDRE} &= \frac{B8A - B5}{B8A + B5}, & \text{NIRv} &= B8 \times \text{NDVI}, \\
 \text{EVI} &= 2.5 \frac{B8 - B4}{B8 + 6B4 - 7.5B2 + 1}.
 \end{aligned} \tag{5.5}$$

where B2, B4, B5, B8, B8A, B11 are reflectance bands. The reflectance values were divided by 10,000 before the spectral indices calculations were done and any reflectance

value of -10,000 was treated as no-data. A small denominator tolerance prevented division by values close to zero for EVI, therefore EVI values outside the interval -1 to 1 were removed because the value can become extremely large when its denominator approaches zero. For the CNN models, these index calculations produced spatial channels for the input chip image rather than footprint averages. The same index definitions were also used for the polygon means experiments. Missing raster pixels were kept as NaN rather than being replaced by a meaningful number. This would help simplify the data normalization step where the predictor means and standard deviations were calculated from the finite training pixels only. After normalization, any remaining missing (NaN) pixels were replaced by zero, which represents the training-channel mean in the normalized space. This prevented raw zero from being confused with valid zero FAPAR, spectral index, or crop fraction values.

The crop data had 21 unique crop labels and therefore had to be simplified into few crop channels that represented the main crop categories in the study area. The channels for the input chip were winter wheat, winter barley, winter rye, maize, grass and forage, woody crops, other crops, active crops, and non-crop land. The woody channel combined vineyards with fruit trees and other woody agricultural vegetation, the grass and forage channel combined the grassland and fodder crop classes. Active-crop fraction was made to signify which crop areas had active crop growth during a particular period and therefore its construction depended on both crop class and observation month. For example, winter wheat's main growth period was considered from February to July and again in October and November (next cycle of crops), while maize was active from May to October. The active crop channel was formed by summing the fractions of crop classes whose predefined growth period included the current month. With this active crop channel, we could present crop data in such a way that it represents the seasonal effect of crop growth. The annual crop map remained year specific so that rotations and changes between years were not replaced by one static landcover map.

The month predictor variable (obtained from the SIF observation date) was represented by two spatially constant channels rather than by integers 2 to 7 (February to July, the month interval for this study). For month m , the channels were calculated as

$$m_{\sin} = \sin\left(\frac{2\pi(m-1)}{12}\right), \quad m_{\cos} = \cos\left(\frac{2\pi(m-1)}{12}\right). \quad (5.6)$$

This representation places December and January next to each other on a cycle/circle and avoids an artificial large distance between month 12 and month 1, as the characteristics of December is similar January. The main CNN model input chip (using Sentinel-2 as the base dataset) therefore contained 19 channels: five spectral indices, FAPAR, PAR, APAR, seven grouped crop fractions, active-crop fraction, non-crop fraction, and the two month channels. Spatial information in the form of Latitude and longitude were not included as image channels as the SIF data was extremely spatially sparse in a given year and therefore during the external inference testing stage there could have been test samples whose Latitude and Longitude were outside the training range for these two variables. Also excluding them encouraged the model to use environmental and seasonal information rather than directly memorizing OCO-2 SIF track locations (each SIF observation for a date is part of long date track as can be seen in FILLREF). During model training, the channel order was written to the dataset configuration and later checked against the model checkpoint during inference. This was important because changing channel order would silently associate trained weights with the wrong predictor.

5.3.4 Model-Ready Data Representations

The processed predictor variables were represented differently for tabular, area-to-point and CNN models. For the polygon mean model (tabular data), every SIF observation becomes one tabular row and the mean of predictor channel c was calculated from the pixels intersecting its footprint:

$$\bar{x}_{i,c} = \frac{\sum_{p \in i} a_{i,p} x_{p,c}}{\sum_{p \in i} a_{i,p}}, \quad (5.7)$$

where $a_{i,p}$ is the fractional area of pixel p inside footprint i . Each row in the data table contained one combined SIF target and one value for each spectral index, radiation and crop predictor, each value representing the mean value of the predictor values falling in one SIF footprint. APAR was calculated as mean FAPAR multiplied by mean PAR. The Random Forest and XGBoost models used this table which removed within footprint spatial structure (as the predictors were reduced to one mean value). The single footprint CNN model datasets centred one multi-channel raster chip on each SIF footprint and stored the SIF footprint as a fractional mask. The predicted SIF map (at predictor resolution) was averaged over this mask before comparison with the observed footprint value. The area-to-point MLP model data represented each footprint as a set of no more than 200 spatially distributed Sentinel predictor rows and assigned the same footprint level 740 nm target to those rows. Each OCO-2 footprint at 1km² contains as many as 2,500 Sentinel-2 pixels (20m). For the sake computational reduction, at most 200 pixels were sampled across the footprint, and their predictor values were given to the MLP. Because no pixel-level SIF observations existed, every sampled pixel shared the footprint's observed 740 nm SIF value. Its pixel level predictions were then averaged back to the footprint level for comparison to original observation value because real observed 20m SIF labels are unavailable (the best resolution for validation is OCO-2 SIF at 1 km²). Therefore, the area-to-point MLP and MODIS 16 by 16 dataset differed from the principal models in both target wavelength (740nm vs combined SIF target) and filtering.

The Multi-footprint (MultiSIF) model datasets grouped several nearby same-date observations in one predictor chip by the means of density clustering keeping a minimum of 4 SIF polygons and max range 10-11 polygons falling in one window, while retaining a separate mask and target for every footprint. The Sentinel-2 version used a 6km by 6km chip at 20m resolution, containing 4-10 footprints and each footprint having at least 90% of its area falling inside the chip. Its errors were averaged within each chip so that chips containing more footprints did not automatically have greater influence. The Spatial aggregate (4km) model dataset used a different target by assigning same-tile, same-date, and same-mode footprints to fixed 4 km cells containing at least five observations. Each cell had 200 by 200 predictor pixels, and its observed target was

$$\bar{S}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} S_i, \quad (5.8)$$

where n_g is the number of assigned footprints and S_i is their observed SIF. For simplicity sake and to keep enough input chips for training purpose, the aggregate weight map gave equal influence to the normalized footprint masks even when their areas inside the chip differed. Since multiple footprints were being averaged down into one value, partial footprint coverage over the chip would not significantly change the accuracy of the overall mean. After this best model (Spatial aggregate (4km)) was selected for further validation, native-footprint inference chips were prepared for two unseen MGRS tiles using the same 4 km context, 20 m grid, and channel order. The resulting 20 m predictions were averaged

over each observed footprint mask and compared to the original SIF observation value. It should be noted that these external samples were not used to refit the model or its calibration, they were only used to see if the model's strong training performance could be carried over to external data.

For the yield model dataset, Bavarian NUTS 3 regions were chosen in the selected five MGRS tiles if at least 50% of their area fell inside an available MGRS tile (Sentinel-2 data is represented in MGRS format), any duplicate tile assignments were removed, one tile was assigned to one NUTS 3 region depending on area overlap. The annual crop type rasters provided the winter wheat pixels spatial location for every retained NUTS 3 region, year and month from March to July. Since the CNN predictor grid was at 20m resolution, wheat map had to be converted to 20m resolution by required all four contributing 10 m crop pixels to be classified as winter wheat. The best model Spatial Aggregate (4km) was used to predict the high resolution SIF map only on the blocks containing these 20m wheat pixels and finally the wheat pixel SIF values were averaged to one crop pure monthly value for each NUTS 3 region. Raw comparison variables were formed from the OCO-2 SIF footprints and from subsets with at least 5%, 10%, or 30% winter-wheat coverage (eg; a SIF footprint has at least 30% of it's crop area as winter wheat). Footprints were first averaged within the acquisition date and then across the available dates in each month so that dense OCO-2 tracks did not dominate the regional predictor (some date tracks, have more SIF observations than other date tracks). The crop-pure and raw monthly predictor values were reshaped into one row per region and year and joined to official winter wheat yield table. The winter wheat yield table was formed by combining yearly NUTS 3 categorised yield files into one table. The yield unit was converted from decitonnes per hectare to tonnes per hectare so that it could be compared with external studies, and rows without a winter wheat yield (target value) were removed. Missing predictor values were replaced using means calculated from 2019-2022 only, so the held-out 2024 observations did not affect preprocessing. The processed table containing the SIF month wise averages and yield value allowed crop-pure enhanced SIF and mixed raw OCO-2 SIF to be compared using the same regional yield records.

5.4 Model Architectures

This section will only discuss the model architectures for the deep learning model experiments. The area-to-point model used a fully connected multilayer perceptron (MLP) that received one 20m predictor row at a time. Each row contained 17 features: Sentinel-2 spectral indices, crop indicators and two cyclical month variables. The input layer was connected to a hidden layer with 64 neurons, followed by a rectified linear unit (ReLU) activation and dropout with a probability of 0.1. A second hidden layer reduced the representation from 64 to 32 neurons and again applied the activation ReLU and dropout. The final linear layer of the MLP model converted the 32 neuron values into one pixel-level SIF estimate. No output activation was used because SIF was treated as a continuous regression target that could include values below zero. The model's MLP weights were applied to every sampled pixel independently, it did not know where the pixel was located relative to other pixels in the footprint or what its neighbouring pixels contained (ie; it has no spatial understanding of how the predicted SIF values are connected in a SIF footprint). Because OCO-2 provides one observed SIF value for the whole footprint rather than separate 20m observations, the model's pixel-level predictions inside that footprint were averaged to one predicted value. This predicted value was compared with the observed footprint level SIF during training to calculate the necessary performance statistics. Therefore, the model learned pixel-level relationships indirectly from footprint level supervision.

The single footprint and multi footprint CNN experiments used a compact U-Net architecture that preserved the spatial arrangement of the predictor channels. The models using MODIS variables (GLASS products) used two encoder levels with 32 and 64 feature maps, followed by a 128 feature map layer (the deepest representation layer of the model) and this was followed by two decoder levels with 64 and 32 feature maps, matching symmetrically. Each convolutional block contained two 3×3 convolutions, and every convolution was followed by batch normalization and a ReLU activation. Likewise the Sentinel-2 multi footprint CNN model (MultiSIF) used the same two-level layout as in single footprint model (MODIS) but started with 16 feature maps, 32 feature maps, and finally 64 feature maps in the deepest layer. Its convolutional blocks used group normalization and the SiLU activation instead of batch normalization and ReLU. This is because the MultiSIF model using Sentinel-2 predictors used small training batches and the 20m resolution image chips were much larger than the MODIS chips and therefore consumed more GPU memory. Batch normalization can become unstable with small batches because it estimates statistics from the current batch whereas group normalization calculates statistics within groups of the feature channels and is therefore less dependent on batch size. SiLU was used because it provided a smooth nonlinear activation and retained small negative responses instead of discarding them completely. In both experiments, 2×2 max pooling reduced the spatial dimensions in the encoder section and 2×2 transposed convolutions restored them in the decoder section. Skip connections concatenated the encoder features with the corresponding decoder features which allowed local boundaries and fine spatial patterns to be recovered back after pooling (pooling would condense these fine spatial patterns). Finally, a final 1×1 convolution produced one SIF value for every output pixel without applying an output activation. The predicted map was then reduced over either one footprint mask (single footprint model) or several separate footprint masks (multi footprint model).

The best model; Spatial aggregate (4km) used a deeper three-level U-Net compared to a two-level U-Net because its input contained 200 pixels by 200 pixels and could support an additional stage of spatial abstraction without losing too much information. A 200 by 200 output would go from 200, 100, 50, 25 after three pooling operations. The deepest layer still has a meaningful 25 by 25 spatial representation, whereas for the smaller 16 pixels by 16 pixels (single footprint) and 24 pixels by 24 pixels (multi footprint), three pooling stages would greatly compress any meaningful information away. The experiment's 19 predictor channels entered the encoder sections with 16, 32, 64 feature maps and final representation layer with 128 feature maps, the decoder section will reverse this sequence and finally a 1×1 output layer restored a single 200 by 200 predicted SIF map. Every block again used two padded 3×3 convolutions with group normalization and SiLU activations. Three max-pooling operations reduced the 200 pixels by 200 pixels input successively to 100 by 100, 50 by 50, and 25 by 25 pixels. The model produced a separate predicted SIF value for every 20m pixel in the output map and during training, the 20 m predictions were combined using the footprint-weight map and compared with the single aggregate target (4km). It should be noted that the 20m outputs are model derived predictions constrained by aggregate observations, rather than independently validated 20m measurements (since real 20m SIF observations do not exist).

5.5 Model Training and Experimental Design

Partitioning the model data into train, validation and tests was done before any data normalization, to avoid train statistics from leaking into the validation and test splits. Each model had different methods of partitioning the data (can call this as the partitioning or split unit). Different methods were chosen to see how it affects model performance with

the final partitioning unit being used for the Spatial aggregate (4km) model. For the best model; Spatial aggregate (4km), input chips that shared either an OCO-2 acquisition date or a Sentinel-2 product were connected into one component, and these complete components were assigned uniquely to the training, validation or testing splits. This helped prevent the same OCO-2 acquisition date observations or predictor image (chips) from appearing in more than one partition. The components were allocated to an approximate 80/10/10 split by number of chips counts (not exact as it should satisfy the following conditions) while balancing month, measurement mode and MGRS tile composition across the three partitions. The Sentinel-2 multi footprint (MultiSIF) model also used a 80/10/10 split but its data partitioning unit was the complete acquisition date and balancing was done on month and measurement mode, ie; the data partitions contain a balanced distribution of the months and measurement modes. Unlike the Spatial aggregate (4km) model, this date based split did not prevent the same Sentinel-2 product from occurring in different partitions, so generalisation over MGRS tiles or regions cannot be ascertained fully with this experiment. The MODIS multi footprint model used year wise partitioning, 2019-2022 subset was used for training, 2023 for validation and 2024 for testing. Whereas the MODIS single footprint models used a 70/15/15 random stratification split by state, year and MODIS composite day. The area-to-point MLP model used 70/15/15 split and ensured that all sampled pixels from one SIF footprint remained together in one split. The polygon mean tree models used a date-grouped 80/20 train/test split stratified by month, measurement mode while the yield models used 2019–2022 region-year subset for training and 2024 subset as the test set. All these different methods of data partitioning were used to identify on a base level the ideal data partitioning technique.

Normalization was done for the predictor variables, and was calculated from the training partition only and then applied unchanged to validation and test data. A random seed of 42 was used for the data partitions and model training procedures. Raster channels were standardized separately, missing raster values were replaced by zero only after standardization as previously mentioned. Fractional footprint and aggregate masks were converted to floating-point values and normalized to sum to one before they were used to reduce a predicted SIF map to its supervised scalar value. For the CNN models, the SIF target were also standardized during optimization and then returned to the original units for evaluation using the following formula

$$SIF_{\text{standardized},i} = \frac{SIF_i - \mu_{\text{train}}}{\sigma_{\text{train}}}, \quad (5.9)$$

where μ_{train} and σ_{train} are the mean and standard deviation of the training targets. This transformation centred the targets near zero and gave them a standard deviation close to one, this helped support a more stable neural-network optimization. After inference, the predicted values were returned to the original SIF units using

$$\widehat{SIF}_i = \widehat{SIF}_{\text{standardized},i} \sigma_{\text{train}} + \mu_{\text{train}}. \quad (5.10)$$

It should be noted that the evaluation metrics such as RMSE and MAE were calculated in the original SIF units. The area-to-point MLP in contrast to the CNN models kept its target in the original SIF units. This was a design choice. Since CNN models contain more parameters than the MLP model, standardizing the SIF targets kept the loss and gradients on a consistent scale, and thus made optimization more stable and allowed the same learning settings to work across the various CNN model experiments. The CNN models and MLP model were optimized with AdamW using an initial learning rate of 10^{-3} and a weight decay of 10^{-4} . The CNN models using Sentinel-2 data processes four separate batches of eight chips (each chip is an image with multiple channels), and after each batch

is done it calculates the gradients but doesn't update the model weights immediately. Only after all four batches are done are the gradients added together and finally the optimizer updates the weights. This method allows the model to be trained with a larger batch than what the GPU memory can permit, ie; the physical batch size is 8, while the effective batch size is 32. For a chip containing several footprints, the footprint losses were first averaged within the chip and then the chip losses were averaged across the batch. This helped prevent the chips containing more footprints from influencing the model training more greatly than chips containing less number of footprints, ie; avoid footprint count in a chip from influencing the model training performance. Huber loss was used for the models using combined SIF target (`target_modis_sif`) and for area-to-point, while mean squared error was used for the 740nm baseline model. Huber loss was used to stop extreme outliers from affecting the model training greatly. The Random Forest and XGBoost models were trained with fixed configuration settings (min tree depth, min node leaves) so that the comparison focused on the input representation and model family rather than effective hyperparameter search.

All the deep learning network models final model checkpoint were selected using the lowest validation RMSE. Models using Sentinel-2 data used early stopping with patience of seven epochs and also reduced the learning rate of the model when validation RMSE stopped improving. The area-to-point MLP allowed training to up to 150 epochs with patience of 20. The MODIS based models used 30 epochs and retained the checkpoint with the lowest validation RMSE. For most of the CNN models, a linear calibration of observed SIF versus predicted SIF was fitted on the validation predictions only. This was then applied unchanged to the test predictions. Both raw and calibrated metrics were kept so that any improvement caused by this post training adjustment remained visible, see FILLREF to see which models chose the calibrated results. The external native footprint inference used the saved Spatial aggregate (4km) model, its training normalization statistics, and its validation derived calibration to centred 4 km windows from two previously unseen MGRS tiles. These 4km windows had 20m SIF maps (4km cause each input chip is 200 pixels by 200 pixels, each pixel 20m resolution) which were averaged over the individual OCO-2 footprint masks and compared with their observed footprint SIF values. Finally the winter wheat yield model used a fixed Random Forest configuration for all crop pure and raw SIF predictor subsets, so that the 2024 test data comparison between the various yield models reflected more of the input data quality rather than any hyperparameter tuning efforts. A more concise view of the experimental settings can be seen in Table 5.1.

5.6 Experimental Environment and Reproducibility

For this study, all data acquisition, preprocessing, spatial matching, raster extraction and model ready data preparation were performed on a local workstation with a 20-core CPU, 32 GB RAM and RTX5070 GPU. For the python processing scripts, Python 3.11.9 in the Positron development environment was used. Python libraries like NumPy and pandas were used for tabular operations, Rasterio, GeoPandas, Shapely and pyproj were used for raster, vector, geometry and coordinate reference system (CRS) operations. For the R processing scripts, R 4.5.2 in the RStudio development environment was used. R libraries, terra and sf provided the main raster and vector functions, the future, furrr, and tidyverse packages provided the necessary parallel and tabular workflow functions. Processing operations that were computationally expensive were made easier to compute by grouping observations by date, product (or tile), cropping rasters before extraction and processing independent groups in parallel where possible. Since the local workstation had 32GB RAM and each Sentinel-2 reflectance band size can vary from 53MB to 230MB, crop

type rasters around 450MB, without doing any kind of grouping, cropping the memory costs can easily balloon to over 32GB RAM. Eventhough some serial workflows for the processing could be done in memory with the ungrouped data, to make use of parallel computing, a smaller sized data needed to be made such that it could be fed to each core of the parallel process. All Neural Network training and evaluation were conducted in Kaggle environments which had two NVIDIA T4 GPUs, a four-core Intel Xeon CPU allocation, and 30 GB of RAM, the provided GPUs were better the local workstation's RTX5070 for model training. The Kaggle models used PyTorch 2.10.0 with CUDA 12.8. Training experiment settings like random seeds, data-partition definitions, normalization statistics, chip metadata, prediction tables, and evaluation outputs were stored with the corresponding jupyter notebooks so that each experiment could be redone without regenerating new partitions, normalization statistics for future steps. The best model; Spatial aggregate (4km) model weights, checkpoint, normalization statistics, and channel-order metadata are provided in the following google drive link [FILLREF](#).

5.7 Evaluation Metrics Application

This section will describe the performance metrics that were used to evaluate the models and also Huber loss function that was used for the main CNN models training loss function. The residual as shown in [FILLREF](#) twice can be described as $e_i = \hat{y}_i - y_i$ where y_i denotes the observed SIF or yield value for test sample i and $\hat{y}_i$ denotes the predicted SIF value or yield value. Residual shows the prediction error for one observation. Positive residual indicates the model over predicted likewise negative value signifies under prediction. Similarly bias error indicates the mean of all residuals and therefore positive and negative values can use the same interpretation. An important point positive and negative residuals can cancel each other, so even if individual errors are large, bias can be near zero. Root mean squared error (RMSE), mean absolute error (MAE), and mean bias error described the predictive error while the coefficient of determination (R^2) measured the fraction of observed variation explained by the predictions. An R^2 value near one indicates that the model explains a large proportion of the variation in the observed values, while a value below zero means that the predictions perform worse than using the observed mean as a constant estimate ie; as a mean baseline model. Since RMSE depends on the spread of the target, normalized RMSE (NRMSE) was also used when comparing targets at different spatial supports, specifically for the Spatial aggregate model (4km) and multi footprint Sentinel-2 model (MultiSIF) comparison. These model metrics were calculated as

$$\begin{aligned}
 \text{RMSE} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}, \\
 \text{MAE} &= \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|, \\
 \text{Bias} &= \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i), \\
 R^2 &= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \\
 \text{NRMSE} &= \frac{\text{RMSE}}{s_y}.
 \end{aligned} \tag{5.11}$$

Huber loss was used the optimization for the main CNN models and the area-to-point

MLP model, with the exact transition parameter varying experiment to experiment as shown in Table 5.1. Huber loss behaves quadratically for small residuals but behaves linearly (linear penalty) for large residuals. This makes model fitting less sensitive to unusually large SIF errors than the mean squared error. The loss can be defined as

$$L_{\delta}(e_i) = \begin{cases} \frac{1}{2}e_i^2, & \text{if } |e_i| \leq \delta, \\ \delta \left(|e_i| - \frac{1}{2}\delta \right), & \text{if } |e_i| > \delta. \end{cases} \quad (5.12)$$

where $e_i = \hat{y}_i - y_i$ is the residual and δ is the transition parameter. Huber loss was used to update the model parameters during training (reduces this loss during training) whereas the metrics RMSE, MAE, bias, R^2 were used to report predictive performance metrics on the validation and test data sets.

Model performance was also further examined using the following metrics: slope and intercept of the least-squares regression of predicted values on observation values, Pearson's correlation coefficient, and distribution of residuals as seen in FILLREF. An ideal model would have a slope near 1, an intercept near 0, a correlation near 1, and residuals narrowly centred around 0. Pearson correlation measures how strongly predicted and observed values change together.

The regression slope b_1 , intercept of the regression line b_0 , Pearson correlation r were calculated as

$$\begin{aligned} b_1 &= \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sum_{i=1}^n (y_i - \bar{y})^2}, \\ b_0 &= \bar{\hat{y}} - b_1 \bar{y}, \\ r &= \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}}. \end{aligned} \quad (5.13)$$

An ideal model has $b_0 = 0$, $b_1 = 1$, zero bias, and small residual spread. Residual stability was also described by its standard deviation and interquartile range ($Q_{0.75}(e) - Q_{0.25}(e)$) and reported for Spatial aggregate (4km) and MultiSIF Sentinel-2 model, this described the spread and stability of prediction errors. For the Spatial aggregate (4km) model, predictor importance was estimated by independently permuting each validation channel (each channel of the image chip in the validation set) and checking the increase in RMSE $I_j = \text{RMSE}_{\text{perm}(j)} - \text{RMSE}_{\text{base}}$, with larger positive values indicating greater predictive importance. The core performance metrics as shown above were calculated for SIF value bins, footprint counts, measurement modes, months, years, MGRS tiles, states, and plant hardiness zones to identify differences in model performance among data groups. Relative RMSE reduction of crop-pure SIF against a raw-SIF baseline was calculated as $100(\text{RMSE}_{\text{raw}} - \text{RMSE}_{\text{crop}})/\text{RMSE}_{\text{raw}}$ and was used for comparison of the different yield models. Finally, given the small number of samples in the held-out 2024 test set, paired bootstrap resampling by region was used to quantify uncertainty in the RMSE difference between the crop-pure and raw-SIF yield models. In each bootstrap repetition, the same test regions were sampled with replacement for both models. This allowed for paired comparison. The 2.5th and 97.5th percentiles of the relative RMSE reduction distribution were used to report the 95% confidence interval.

Table 5.1: Training and partition settings used in the modelling experiments.

Experiment	Split Unit	Loss	Optimizer	Batch	Epochs/trees	LR	ES	Calibration
4 km aggregate U-Net	Component 80/10/10	Huber (1.0)	AdamW	8×4	30	10^{-3}	7	Yes
Sentinel-2 multi-SIF U-Net	Date 80/10/10	Huber (1.0)	AdamW	8×4	30	10^{-3}	7	Yes
MODIS multi-SIF U-Net	Year 2019-22/23/24	Huber (0.5)	AdamW	128	15	10^{-3}	-	Yes
MODIS single-SIF U-Net	Stratified 70/15/15	Huber (0.5)	AdamW	256	30	10^{-3}	-	Yes
MODIS 740 nm baseline	Stratified 70/15/15	MSE	AdamW	512	30	10^{-3}	-	No
Area-to-point MLP	Footprint 70/15/15	Huber (0.1)	AdamW	-	150	10^{-3}	20	No
Polygon means: XGBoost	Date 80/20	Squared error	XGBoost	-	800	0.03	-	No
Polygon means: Random Forest	Date 80/20	Squared error	RF	-	700	-	-	No
Winter-wheat yield RF	Year 2019-22/24	Squared error	RF	-	500	-	-	No

6 Results and Discussions

6.1 Experimental Overview

This section will give an overview of the experiments done in this study and expected result. The first stage evaluated how accurately OCO-2 SIF could be predicted from satellite derived spectral, biophysical, radiation, seasonal and crop composition under several model, input data representations and target representations. The evaluated models include polygon mean Random Forest and XGBoost regression, area-to-point multilayer perceptron, single footprint U-Net (MODIS predictors), multi footprint U-Net (MODIS and Sentinel-2 predictors) and the principal Spatial aggregate (4km) U-Net. The spatial supports (data representation) ranged from tabular footprint level summaries to spatially structured image chips (with multiple channels) and therefore tested both model architecture & choice and how the predictor information was represented. The initial baseline models used target SIF at 740nm before moving on to the uncertainty corrected combined SIF target derived from SIF at 757nm and 771nm as the later was found to increase the signal-to-noise ratio and thus make any SIF predictions more reliable. Training, validation, and test data were partitioned according to each dataset's temporal and spatial structure, complete dates, footprints, or connected date-product components kept together where used as split units, as described in Table 5.1. The model performance was assessed using RMSE, MAE, bias, R^2 , regression slope, residual behaviour, data subgroup metrics and observed-versus-predicted SIF comparisons.

The results answer more general SIF prediction performance to more specific questions about data representation, model generalizability & stability, spatial variation and agricultural utility. The best model; Spatial aggregate (4km) was internally evaluated on held out chips (image with channels) and then externally evaluated on two previously unseen MGRS tiles on 4km windows (to match with input data extent) which produced 20m SIF maps which were then aggregated for each OCO-2 footprint and compared to original SIF observation. This external testing allowed to confirm if 20m SIF predictions aggregated to OCO-2 footprint resolution (1km^2) was viable. If confirmed, it would indicate the model is able to at least make SIF predictions at 1km^2 resolution and thus tackle the task of spatially and temporally completing a SIF map at OCO-2 SIF resolution. Comparisons between tree based, MLP and CNN models show how tabular summaries and spatial predictor grids affected accuracy of the SIF predictions. The effect of spatial aggregation on SIF prediction and retrieval-noise reduction is examined by comparing native footprint (MultiSIF Sentinel-2 model) and 4km targets (Spatial aggregate 4km model), their observed vs predicted SIF values variability, residual distributions, and performance across different footprint counts. Model behaviour is also evaluated across data subgroups for the following predictor variables: months, measurement modes, years and plant-hardiness zones which then helps identify conditions associated with stronger or weaker predictions. Finally the application of high resolution SIF predictions in winter wheat yield models were evaluated. Crop pure enhanced SIF was compared with raw mixed-footprint OCO-2 SIF at varying winter wheat coverages for select Bavarian NUTS-3 regions, testing was done on the held out 2024 subset. These results would then be compared with existing SIF enhancement approaches.

Table 6.1: Calibrated model performance by observed SIF interval for the combined nine-tile dataset, the held-out nine-tile test set, and the external native-footprint evaluation over two additional tiles.

A. Combined training, validation and test data: nine tiles								
SIF interval	n	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2	Pearson r
$[-0.25, 0)$	66	-0.0246	0.0851	0.1264	0.1097	0.1097	-32.616	-
$[0, 0.25)$	2,920	0.1373	0.1560	0.0645	0.0466	0.0187	-0.028	-
$[0.25, 0.50)$	3,846	0.3721	0.3631	0.0662	0.0520	-0.0090	0.091	-
$[0.50, 0.75)$	1,076	0.5712	0.4718	0.1187	0.1008	-0.0995	-3.380	-
$[0.75, 1.00)$	12	0.7873	0.5149	0.2756	0.2724	-0.2724	-56.094	-
B. Held-out test data: nine tiles								
SIF interval	n	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2	Pearson r
$[0, 0.25)$	274	0.1509	0.1761	0.0658	0.0437	0.0251	-0.247	-
$[0.25, 0.50)$	545	0.3636	0.3455	0.0605	0.0479	-0.0181	0.177	-
$[0.50, 0.75)$	112	0.5643	0.4667	0.1119	0.0983	-0.0976	-3.810	-
C. External native-footprint evaluation: two additional tiles								
SIF interval	n	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2	Pearson r
$[-0.5, -0.25)$	4	-0.2689	0.3181	0.5876	0.5870	0.5870	-3884.368	0.999
$[-0.25, 0)$	104	-0.0625	0.1816	0.2798	0.2441	0.2441	-25.084	-0.055
$[0, 0.25)$	652	0.1413	0.2249	0.1534	0.1174	0.0836	-4.232	0.274
$[0.25, 0.50)$	827	0.3707	0.3617	0.1104	0.0893	-0.0091	-1.375	0.307
$[0.50, 0.75)$	350	0.5942	0.4319	0.1864	0.1634	-0.1623	-6.672	0.184
$[0.75, 1.00)$	59	0.8195	0.4491	0.3832	0.3704	-0.3704	-44.830	-0.141
$[1.00, 1.25)$	4	1.1151	0.3941	0.7233	0.7210	-0.7210	-213.446	0.324

Table 6.2: Calibrated fitted performance across measurement mode, temporal alignment, month, and observation-year subgroups.

A. Measurement mode								
Mode	Obs. min	Obs. max	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2
Mode 0	0.0306	0.7200	0.3105	0.3030	0.0659	0.0492	-0.0075	0.8090
Mode 1	0.0131	0.6790	0.3412	0.3181	0.0746	0.0566	-0.0231	0.6885
B. Date alignment								
Alignment	Obs. min	Obs. max	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2
Within range	0.0131	0.7200	0.3379	0.3219	0.0712	0.0539	-0.0160	0.7421
Outside range	0.0570	0.4855	0.1820	0.1794	0.0569	0.0393	-0.0026	0.6428
C. Observation month								
Month	Obs. min	Obs. max	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2
February (2)	0.0285	0.1449	0.0869	0.0861	0.0302	0.0236	-0.0008	-0.2443
March (3)	0.0452	0.3008	0.1267	0.1172	0.0343	0.0263	-0.0095	0.4035
April (4)	0.0131	0.5578	0.2999	0.2890	0.0582	0.0453	-0.0109	0.5123
May (5)	0.0828	0.7200	0.4493	0.4193	0.0876	0.0712	-0.0301	0.4355
June (6)	0.2119	0.6527	0.4200	0.4035	0.1025	0.0753	-0.0165	-0.2676
July (7)	0.0441	0.5177	0.3447	0.3463	0.0816	0.0609	0.0016	0.2947
D. SIF observation year								
Year	Obs. min	Obs. max	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2
2019	0.0131	0.6527	0.3117	0.3021	0.0654	0.0493	-0.0095	0.4809
2020	0.0452	0.3008	0.1267	0.1172	0.0343	0.0263	-0.0095	0.4035
2022	0.0828	0.7200	0.4482	0.4190	0.0890	0.0721	-0.0293	0.4202
2023	0.2315	0.5481	0.4099	0.4069	0.0729	0.0579	-0.0030	0.0938
2024	0.0285	0.1449	0.0869	0.0861	0.0302	0.0236	-0.0008	-0.2443

6.2 Overall SIF Prediction Performance

The primary model for this study; Spatial aggregate (4km) U-Net was evaluated first on held out 4km grid cells from the nine MGRS tiles (internal dataset for training, validation, testing) and later evaluated on native OCO-2 footprints from two additional MGRS tiles. This testing process allowed to see how the performance changed when the spatial support was changed from 4km grid to the noisier and finer native footprint support, remember the spatial aggregation was done primary to reduce the retrieval noise during data acquisition. The calibrated model reported RMSE of 0.0701 and R^2 of 0.761 for 931 grid cell targets for the held out 4km test set as shown in Figure ???. The predictions regression on observations had a slope of 0.726 and an intercept of 0.074. Therefore the model was able to explain a substantial part of the variation in the aggregated SIF target. The slope below one shows that its predicted range was narrower than the observed range, ie; low SIF values were overpredicted while the largest values were generally underestimated. One reason could be the lower sample size of these values as seen in the combined fitted SIF bin performance table FILLREF. In light of this, the strong spatial aggregate result should be interpreted as accurate prediction of the noise reduced 4km SIF means rather than a direct validation for every 20m output SIF pixel. To verify if this strong performance can be carried out to a much lower resolution, it was validated with the lowest resolution real SIF data (OCO-2 SIF footprints).

The external native footprint experiment applied the same trained model and validation derived post training calibration to 2,000 individual OCO-2 footprints in the two new MGRS tiles that were not seen during model training. For the 2,000 footprints, the calibrated RMSE was 0.1701, R^2 was 0.373, and regression slope was 0.400 (Figure ??). The mean bias was only -0.0034 , but this small value could be the result of positive and negative residual values cancelling each other out as previously discussed. The external test plot shows strong range compression, with low and negative observations predicted too high and high observations predicted too low. Given that our SIF range seen during model training was around $[0, 0.75)$, 1,829 external footprints evaluation with SIF range $[0, 0.75)$ reduced RMSE to 0.1424 and increased slope to 0.472 while R^2 remained 0.349 (Figure ??). The lower RMSE is expected since the reduced external SIF range matches closely to the training SIF range. No change in R^2 can attributed to the native SIF samples being extremely noisy as can be seen in (Figure ??), but high density of points can still be seen closer to the diagonal.

The interval-specific results in Table 6.1 can clarify this behaviour. In the held-out nine tile test set, RMSE was 0.0658 for SIF in $[0, 0.25)$ and 0.0605 for $[0.25, 0.50)$, but increased to 0.1119 for $[0.50, 0.75)$. The corresponding bias changed from $+0.0251$ in the lowest interval to -0.0976 in the highest interval, thereby confirming the tendency of the model to pull predictions towards the centre of the target distribution. This pulling effect was stronger at native footprint scale where RMSE decreased from 0.1534 in $[0.0, 0.25)$ to 0.1104 in $[0.25, 0.50)$ and increased to 0.1864 in $[0.50, 0.75)$ and 0.3832 in $[0.75, 1.00)$. The very small sample counts in the extreme intervals mean that their individual statistics are uncertain, but the overall direction of the errors is consistent with the observed-versus-predicted plots. We can see that model was able to predict well for native OCO-2 SIF resolution for the SIF range $[0.0, 0.75)$. It should be noted that negative within-bin R^2 values should not be read as a contradiction of the overall test R^2 because each narrow SIF interval contains much less target variation, so even moderate absolute errors can perform worse than that's interval mean predictor. Spatial aggregation largely removed negative SIF samples and therefore it cannot be expected for the model to predict them in the native external set.

The performance also varied across acquisition date, temporal subgroups, measurement

modes (Table 6.2). Measurement mode 0 (Nadir) subgroups had an RMSE of 0.0659 and an R^2 of 0.809 versus 0.0746 and 0.689 for mode 1 (Glint), this slight variation is expected as Nadir will capture the signals more strongly than Glint FILLREF. SIF observations whose acquisition dates were within (inrange) the temporal range of their Sentinel-2 products had a higher R^2 than observations outside (outrange) that range (0.742 versus 0.643). The date_align RMSE values were lower for outrange samples because its observed SIF range was much smaller than inrange samples. For the months, we can see the RMSE increasing monthwise till June, this corresponds to the SIF range also being more variable as the months proceeded. Year specific results should be taken with caution as the available months and sample counts differed among years, therefore it does not form a controlled test for generalization among different years.

Feature importance for this model was acquired by permutation importance of the validation set. Figure ?? shows that FAPAR caused the largest increase in validation RMSE when permuted, followed by the seasonal cosine channel and NIRv. Since $SIF \approx APAR \times \Phi_F \times f_{esc} = PAR \times fAPAR \times \Phi_F \times f_{esc}$, SIF and GPP share dependence on absorbed radiation, it makes sense that FAPAR was the most important variable, likewise FILLREF showed NIRv to be important in their SIF enhancement model and the same effect can be seen here too. APAR, NDVI and EVI were the next influential predictor variables and PAR alone, most crop fraction channels had relatively small RMSE improvement effects. The higher importance of APAR than PAR shows that the absorbed radiation supplied more useful information to the model than the incoming radiation. Among the crop channels, the active crop channel contributed more to the RMSE improvement than the individual crop fractions, signifying seasonal representation of crop composition data provided more useful information to the model. It should be noted that these values are conditional permutation results rather than causal effects. To reiterate, it shows how much the trained model relies on each predictor given all the other predictors are available, it does not prove that changing the predictor physically causes the SIF to change. Also another important point, correlated predictors can substitute for one another, therefore any near-zero or negative importance can indicate redundancy of some predictors rather than physical irrelevance.

The spatial residual map for the internal test samples as seen in Figure ?? shows the spatial distribution of over and underestimation of SIF values. Positive and negative residuals occurred across the nine tiles showing that the overall error was composed of spatially varying local outcomes rather than one uniform outcome. This map is useful for identifying locations where differences in land cover, predictor quality, acquisition timing or distribution of the OCO-2 footprints may have affected the prediction of SIF values. It should be noted that this residual map is only for the nine tiles internal testing and doesn't include external native footprint residuals. The terminal colours on the map represent both the clipping threshold and any extreme residuals as the colour scale was clipped symmetrically at 98 percentile ((± 0.180)). The final results for all these figures and tables show that high resolution spectral, biophysical, radiation, seasonal and crop information predicted noise reduced 4km SIF target with good accuracy. This model was able to carry out its performance for the native external set for the SIF range [0.0, 0.75) particularly. Since the scope of this study is only for Germany, acquiring noise 4km SIF samples outside this range may not be possible or may not yield a sizeable sample count for the model to train on [0.75, 1.5) SIF range samples. Finally the predicted 20m SIF maps should be described as spatial downscaling limited to the fine resolution predictors and aggregate OCO-2 supervision, it cannot be validated as 20m SIF values as no real SIF dataset exists at 20m resolution.

6.3 Effects of Model Architecture and Input Representation

Model architecture comparison covered the tree based regressions, area-to-point multilayer perceptron and several U-Net configurations, each using either polygon level tabular predictors or spatial raster chips (images with channels). Table 6.3 shows that the Spatial aggregate (4km) U-Net achieved the strongest internal result, with an RMSE of 0.0701 and an R^2 of 0.761. When tested on an external native OCO-2 footprint resolution, it was able to achieve an RMSE from 0.1534 to 0.1864 for the SIF range $[0.0, 0.75]$, but it was not able to carry out the strong performance outside this range as the internal training dataset SIF range was limited to around $[-0.25, 1]$ with only about 78 samples for the extreme ranges $[-0.25, 0)$ and $[0.75, 0)$. For the following native footprint experiments (single, multi SIF footprints, mean aggregated predictions), the multi footprint Sentinel-2 model obtained an RMSE of 0.1603 and R^2 of 0.418. The MODIS single-footprint model was similarly competitive, with an RMSE of 0.1656 and an R^2 of 0.457. This indicated that both 20m and 250m base predictor resolutions were suitable for SIF predictions. Polygon-means XGBoost and Random Forest produced RMSE values of 0.1650 and 0.1689, with R^2 values of 0.453 and 0.426, respectively. This also showed reducing the predictor values to one single mean value over the target support was competitive with the CNN models that could see internal spatial configurations. The small differences between these native footprint experiments indicate that retaining a spatial structure for the predictors did not automatically improve footprint level predictive accuracy over polygon mean predictors. The MODIS baseline model and the area-to-point MLP used SIF at 740nm as the target variable, they reported much higher RMSE values of 0.2693 and 0.2503 than the models that uses the combined SIF target derived from SIF at 757nm and 771nm. This indicated that the combined SIF target was able to increase the signal-to-noise ratio compared to the SIF at 740nm target. It should be noted that the results compare complete modelling systems that differ in target definition, predictor source, spatial support, calibration, data partition and sample size, therefore it cannot theoretically provide a controlled test of model architecture alone.

The observed-versus-predicted plots for all the native footprint models shown in Figure ?? show a compression of the predicted SIF range toward its centre. We can see the spatial aggregate (4km) model is able to keep the predictions more closer to the diagonal line than the native footprint models ie; the distribution is more closer to the line indicating the noise reduction. The fitted slopes were approximately 0.439 for the multi footprint Sentinel-2 model, 0.444 for Random Forest, and 0.465 for XGBoost, while for the spatial aggregate (4km) model it had a much higher slope of 0.726. We can also see that the spatial aggregate model can predict SIF values better in the $[0.0, 0.75]$ range than the native footprint models, extreme SIF ranges remain a challenge for all the models. Likewise the residual distributions in Figure ?? support the same interpretation: the Spatial aggregate (4km) model had a much narrower residual distribution, while models evaluated against individual footprints showed broader tails. A large part of this difference can be attributed to the noise reduction produced by aggregating several close OCO-2 observations rather than the model architectures. The SIF prediction maps along with equivalent NDVI map in Figure ?? demonstrate an important representation advantage of the image based models as they are able to reflect patterns much more complex because their output preserved field scale spatial patterns and surrounding landscape spatial context. XGBoost and RandomForest (not shown) was also applied to form a 20m SIF predicted map since each predictor variable had a 20m resolution, but it should be noted that each SIF pixel was predicted independently and any visual spatial continuity in the predicted map is from the predictor raster structure rather than any learned neighbourhood relationships. These predicted SIF maps are spatial downscaling products rather than direct pixel level validated

maps (since no true 20m SIF data exists). To summarize the results, target preparation and spatial support (predictor representation) had a larger influence on the accuracy of prediction rather than the choice of the model architectures (tree based, MLP, CNN).

6.4 Spatial Aggregation, Stability and Generalization

Spatial aggregation of SIF observations (nearest samples averaged to one value for one grid cell) greatly reduced the variability of the observed targets and also model errors. The standard deviation (SD) of the observed SIF decreased from 0.210 for multi SIF (multiple sif footprints in one window but treated as individual footprints. We can also call these native footprint) to 0.144 for 4km spatial aggregates, the interquartile range for the SIF value also decreased from 0.291 to 0.203 shown in Table 6.5, signifying SIF range compression as a result of spatial grouping. The residual standard deviation dropped from 0.159 to 0.069 (native footprint to spatial aggregate) and the normalized RMSE improved from 0.763 to 0.488. NRMSE was used for strict comparison purposes to account for the fact that observed SIF SD fell from 0.210 to 0.144, individual model effectiveness will use RMSE. So after taking care of this SD change, the RMSE dropped thereby showing that the spatial aggregate 4km model still performed better even after accounting for the smaller variation in the aggregated SIF values. Figure ?? shows a tighter observed vs predicted relationship for the aggregated model, showing less scatter around the fitted regression line and a slope closer to one. Footprint count analysis as seen in Figure ?? shows additional evidence that spatial aggregation reduced SIF retrieval noise, where the mean target standard error declined from approximately 0.060 for chips (4km windows) containing five or six footprints to about 0.040 for chips containing at least thirteen. Likewise RMSE decreased from about 0.080 for five or six footprints to about 0.062 for eleven or twelve footprints. We can see a slight increase in RMSE for thirteen or more footprints, this could indicate that adding more observations to 4km window may not necessarily guarantee improvement in prediction as high footprint count chips may represent neighbouring cells that are different from each other. Spatial aggregation therefore improved target stability and predictive accuracy but it also narrowed the target range from approximately -0.25 - 1.25 at native support to 0.0 - 0.72 at 4km support and removed most negative and very high SIF observations.

Table 6.1 and Table 6.4 provides the SIF interval level results. For the held out 4km test set (internal), RMSE was 0.066, 0.061, and 0.112 in the $[0, 0.25)$, $[0.25, 0.50)$, and $[0.50, 0.75)$ SIF intervals showing strong performance whereas for the native footprint model, higher RMSE values of 0.136, 0.111, and 0.201 was reported for the same SIF intervals. The Spatial aggregate (4km) model was tested out on an external native footprint test set and showed RMSE values 0.153, 0.110, and 0.186 which was closer to the native multi footprint model. One of the major reason for that is the native footprint is quite noisy inherently, but visual inspection of the external observed vs predicted plot in (Figure ??) shows there is a high density of points close to the diagonal line. The aggregate model's external errors were large for observations outside the aggregate training range, reaching an RMSE of 0.280 for $[-0.25, 0)$, 0.383 for $[0.75, 1.00)$, and 0.723 for $[1.00, 1.25)$ thereby showing that learned relationships in the $[0.0, 0.75)$ range cannot be carried over to the extreme SIF intervals. Violin plots as shown in Figure ?? illustrates the effect of spatial aggregation, where the greater width of the violin indicates a higher concentration of observations. The 4km model's residual distribution is more concentrated around zero compared to the native footprint residuals which have a wider spread and longer tails. This indicates that the aggregation reduced target variability and increased prediction stability. Since the aggregate internal, multi footprint, and aggregate external results were obtained from different spatial supports and test observations, we can only confirm that these com-

Table 6.3: Comparison of the calibration status, SIF target, sample size, and fitted performance of the evaluated RQ2 model configurations.

Model	Calibration	SIF target	n	RMSE	MAE	Bias	R^2	Pearson r
Spatial Aggregate 4 km (SA4km)	Calibrated	MODIS SIF	931	0.0701	0.0527	-0.0149	0.761	-
SA4km OCO-2 native inference	Calibrated	MODIS SIF	2,000	0.1701	0.1300	-0.0034	0.373	0.612
SA4km OCO-2 native, SIF [0, 0.75)	Calibrated	MODIS SIF	1,829	0.1434	0.1135	-0.0054	0.349	0.613
MultiSIF-Sen2	Calibrated	MODIS SIF	6,411	0.1603	0.1223	-0.0166	0.418	-
MODIS 24 × 24	Calibrated	MODIS SIF	2,187	0.1679	0.1293	0.0010	0.349	-
MODIS 16 × 16	Calibrated	MODIS SIF	22,149	0.1656	0.1255	0.0011	0.457	-
MODIS 16 × 16	Uncalibrated	740 nm SIF	26,905	0.2693	0.2107	0.0588	0.350	-
MLP area-to-point	Uncalibrated	740 nm SIF	5,291	0.2503	0.1912	-0.0053	0.415	0.646
PolyMeans: XGBoost	Uncalibrated	MODIS SIF	5,889	0.1650	0.1274	0.0125	0.453	0.675
PolyMeans: Random Forest	Uncalibrated	MODIS SIF	5,889	0.1689	0.1311	0.0146	0.426	0.656

Table 6.4: Calibrated MultiSIF test performance across observed SIF intervals.

SIF interval	n	Obs. min	Obs. max	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2
$[-0.50, -0.25)$	1	-0.2513	-0.2513	-0.2513	0.4294	0.6808	0.6808	0.6808	-
$[-0.25, 0)$	306	-0.2329	-0.0002	-0.0555	0.1689	0.2650	0.2244	0.2244	-25.959
$[0, 0.25)$	2,240	0.0001	0.2499	0.1411	0.2064	0.1360	0.1015	0.0653	-3.013
$[0.25, 0.50)$	2,606	0.2502	0.4997	0.3664	0.3394	0.1105	0.0894	-0.0270	-1.546
$[0.50, 0.75)$	1,063	0.5001	0.7495	0.5943	0.4227	0.2008	0.1743	-0.1716	-7.940
$[0.75, 1.00)$	186	0.7511	0.9889	0.8269	0.4832	0.3567	0.3437	-0.3437	-34.460
$[1.00, 1.25]$	9	1.0281	1.2161	1.0891	0.5101	0.5854	0.5790	-0.5790	-117.500

Table 6.5: Comparison of the spatially aggregated and multi-footprint Sentinel-2 U-Net models. Continuous values are rounded to three decimal places.

Metric	Spatial U-Net	Multi-footprint U-Net
Supervision	Spatially aggregated SIF	Individual SIF footprints within multi-footprint chips
Target support	4 km fixed-grid cell	Native OCO-2 footprint
Predictor resolution	20 m	20 m
Training samples	6,060	4,983
Validation samples	929	553
Test samples	931	686
Test dates	21	18
Observed mean	0.325	0.320
Observed SD	0.144	0.210
Observed IQR	0.203	0.291
Observed minimum	0.013	-0.251
Observed maximum	0.720	1.216
RMSE	0.070	0.160
Normalized RMSE	0.488	0.763
MAE	0.053	0.122
Bias	-0.015	-0.017
R^2	0.761	0.418
Regression slope	0.726	0.439
Regression intercept	0.074	0.163
Residual SD	0.069	0.159

parisons describe general patterns rather than strict paired sample-by-sample patterns. In conclusion, aggregation provided useful SIF retrieval noise reduction, but since the aggregation caused the loss of extreme targets, the model couldn't learn these extreme targets. A larger scope of the dataset outside Germany is needed to get a larger sample size of SIF values in the range [0.75, 1.25).

6.5 Agro-Climatic and Seasonal Variation in Model Performance

The combined fitted evaluation 6.6 uses the training, validation, and test predictions from the Spatial aggregate (4km) to examine inter plant hardiness zone performance given that the overall model performance has been established in the previous sections. Table 6.6 shows that zone 7b had the lowest RMSE at 0.0690, while zone 7a had the highest R^2 at 0.8681. SIF prediction performance was slightly weaker in zones 8a, 8b which had bigger sample sizes, they reported RMSE of 0.0772 and 0.0803 and R^2 0.7812 and 0.7553 respectively. Zone 9a had the highest RMSE of 0.0946 and largest bias of -0.0272 , but this estimate was based on a small sample size of 58 observations obtained from 14 date tracks and two MGRS tiles. The choropleth maps in Figure ?? show the geographical distribution of the differences in RMSE values with respect to plant hardiness zones. It should be noted that the reported values represent broad zone level performance rather than continuous local accuracy, ie; a single value for an individual plant hardiness zone. Figure ?? are the residual boxplots of each available plant hardiness zone and shows similar central residual distributions across the five zones, with residual medians generally below zero and occasional large positive and negative errors. Again, the smaller sample size for zone 9a makes its residual distribution less reliable compared to the other zones.

Table 6.6: Calibrated fitted performance by individual hardiness zone.

HZS	n	Obs. min	Obs. max	Obs. mean	Pred. mean	RMSE	MAE	Bias	R^2	Dates	Tiles
7a	151	-0.0496	0.7544	0.2477	0.2337	0.0703	0.0518	-0.0140	0.8681	30	1
7b	894	-0.0347	0.7462	0.3190	0.3038	0.0690	0.0517	-0.0153	0.8176	63	5
8a	4,458	-0.0937	0.8291	0.3066	0.2983	0.0772	0.0585	-0.0083	0.7812	201	9
8b	1,502	-0.0891	0.8783	0.3155	0.3021	0.0803	0.0608	-0.0134	0.7553	169	8
9a	58	-0.0040	0.7881	0.4114	0.3842	0.0946	0.0697	-0.0272	0.7752	14	2

Figure ?? shows the month versus plant hardiness zone RMSE performance plot, where the RMSE was lowest in February and March with values in 0.034-0.053 range. Errors increased for months April-July, with May and June reporting the highest RMSE values. RMSE reached 0.094 in zone 7a during May, 0.096 in zone 8b during June, and 0.090 in zone 8a during both June and July. There were some combination of zone-month values that count less than 20 or no data, these RMSE values were not reported as the sample size is too low to draw a meaningful conclusion. The increase in RMSE for the later months might be partly due to the expansion of SIF range values (wider SIF variation), changing crop conditions and differences in the spatial distributions rather than a purely climatic zone effect. Across the high sample size zones 7b, 8a, 8b, RMSE varied by only about 0.011, thereby suggesting that the model maintained reasonably stable performance across different agro-climatic conditions. The month to month changes were larger than the differences between these zones, thereby indicating that the seasonal crop development and SIF variability mattered more on the prediction error than plant hardiness zone membership alone. Lastly the negative biases and residual medians indicate a small tendency for the model to under predict SIF across almost all zones.

6.6 Winter-Wheat Yield Prediction

The winter wheat yield experiments compared crop pure enhanced SIF, raw OCO-2 footprints with at least 5%, 10%, 30% wheat coverage and all available raw OCO-2 footprints in five MGRS tiles using a standard fixed configuration Random Forest. A fixed Random Forest configuration was used as the main purpose of this experiment was to see predictive power of the individual data subsets rather than model configurations. It should also be noted that the 10% and 30% wheat coverage subsets were excluded from this result section as they were found to have very few reliable samples owing to the fact most of the SIF footprints had a low wheat farm area due to small farm plots and high mixed crop farms. Table 6.7 shows that the crop pure yield model achieved the lowest 2024 test RMSE at 0.9406 t/ha versus 1.0285 t/ha for the raw 5% wheat coverage model and 1.0874 t/ha for the all footprint model. These differences correspond to RMSE reductions of approximately 8.5% and 13.5% when crop pure SIF was used. The crop pure model also had the lowest MAE, bias, and residual standard deviation, but its normalized RMSE was above one and its R^2 was still negative at -0.1456 but higher than the raw SIF models. Figure ?? shows that all the three models compressed predictions toward the average yield. The crop pure model showed a clearer positive relationship with observed yield than either of the raw SIF models.

Figure ?? shows a mean crop pure RMSE improvement of about 0.084 t/ha over the 5%-wheat model, with an estimated probability of 0.879 that crop pure SIF produced the lower RMSE. It also shows the mean crop pure RMSE improvement of 0.143 t/ha over the all footprint model with a probability of 0.918. However it should be noted that, the 95% bootstrap intervals were $[-0.066, 0.213]$ and $[-0.052, 0.347]$ t/ha with both crossing zero, so the improvement was not statistically conclusive at the 95% level. Paired bootstrap resampling was used because the test set contained only 48 test samples (48 NUTS3 regions). It estimated how uncertain the RMSE difference was while ensuring that both models were repeatedly compared using the same sampled regions. The pair regional comparison as shown in Figure ?? shows an uneven spatial response. Crop pure SIF reduced absolute error in 27 of 48 regions relative to 5% model and 26 out of 48 regions relative to all footprint model. The takeaway is that the results provide preliminary evidence that crop pure enhanced SIF can improve winter wheat yield prediction compared to the raw SIF (mixed crop SIF signals), but the negative R^2 values and wide bootstrap intervals show that the present models are not yet sufficient for making claims regarding strong yield prediction claims. A negative R^2 means the model predicted the 2024 yields worse than a simple baseline that assigns every test region the mean observed 2024 yield. The reasons for this could be the very small test dataset and year to year distribution differences. Therefore the results can only support relative improvement from raw SIF to crop pure SIF, but cannot be treated as a reliable stand-alone yield prediction model. Additional observation years and more NUTS 3 regions would be needed to determine whether the crop pure SIF advantage remains stable in a larger yield dataset.

Table 6.7: Comparison of winter-wheat yield-model performance using three monthly SIF aggregation strategies.

Model	n_{train}	n_{test}	RMSE	NRMSE	MAE	R^2	Bias	Slope	Intercept	Residual SD	Bootstrap RMSE interval
Crop-pure monthly SIF	197	48	0.9406	1.0591	0.7598	-0.1456	0.3507	0.1798	5.8570	0.8820	[0.7679, 1.1068]
Raw $\geq 5\%$ wheat monthly SIF	197	48	1.0285	1.1581	0.7769	-0.3698	0.4395	0.0288	6.9595	0.9397	[0.7737, 1.2612]
Raw all-footprint monthly SIF	197	48	1.0874	1.2244	0.7911	-0.5311	0.4622	-0.0132	7.2645	0.9947	[0.8045, 1.3663]

6.7 Comparison with Existing Techniques and Integrated Discussion

Bring the findings together, compare them with related studies, and discuss Benefits of spatial aggregation Value of high-resolution predictor structure Limitations of native-footprint transfer Agro-climatic variability Usefulness of crop-pure SIF for yield modelling Methodological limitations and practical applicability

7 Conclusion

The final chapter summarizes the thesis. The first subsection outlines the main ideas behind Component X and recapitulates the work steps. Issues that remained unsolved are then described. Finally the potential of the proposed solution and future work is surveyed in an outlook.

7.1 Summary

Explain what you did during the last 6 month on 2 or 3 pages!

The work done can be summarized into the following work steps

- Analysis of available technologies
- Selection of 3 relevant services for implementation
- Design and implementation of X on Windows
- Design and implementation of X on mobile devices
- Documentation based on X
- Evaluation of the proposed solution

7.2 Dissemination

Who uses your component or who will use it? Industry projects, EU projects, open source...? Is it integrated into a larger environment? Did you publish any papers?

7.3 Problems Encountered

Summarize the main problems. How did you solve them? Why didn't you solve them?

7.4 Outlook

Future work will enhance Component X with new services and features that can be used ...

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Annex

```
<?xml version="1.0" encoding="UTF-8"?>
<widget>
    dummy source code
</widget>
```

Listing 1: Sourcecode Listing

dummy text

Listing 2: SIP request and response packet[?]